

Robotics Setter Technician Program

Robotic welding, from learning to improvement

Diego Adair de León Márquez

Universidad Politécnica de Aguascalientes | Mechatronics Engineering, 9th term

Plant Tutor: Fernando Robledo

Engineering Department | August 1, 2026 to January 1, 2027

TENNECO

THE TENNECO VALUE I IDENTIFY WITH

Win

We must earn the trust of our employees and customers.

For me, winning meant taking the longest, hardest weld off a team member's hands.

Program scope and objective

Robotics Setter Technician | August 1, 2026 to January 1, 2027

OBJECTIVE

Become a specialist in robotic welding and advanced technical support.

WHY IT MATTERS

When a welding robot stops, the line stops with it. A setter who can find the cause and adjust the cell on the spot keeps parts moving and quality under control.

FINAL PROJECT

An implemented improvement

In quality, availability or productivity.

Mine: 4M, implemented, with 47% less cycle time.

An updated training process

How the next setters learn.

Mine: study material reviewed, schedule rebuilt.

Four levels, from understanding to improving

The skill path of the program

I

Understands

Safety, the welding process, the procedure (WPS), defects and settings.

L

Does it with support

Basic adjustments, checking settings and changing consumables.

U

Does it alone

Programming the robot (SKS), creating programs and setting them up.

O

Improves and teaches

Troubleshooting, training others, Lean and continuous improvement.

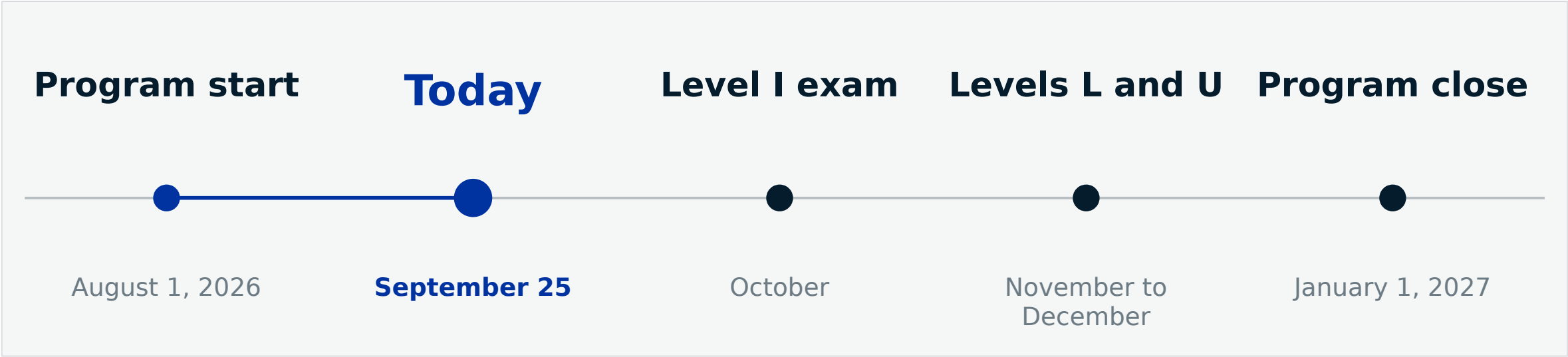
HOW IT WORKS

Each level builds on the one before: first understand, then do, then own it, then improve it.



Eight weeks in, fourteen to go

Program timeline



Level I: understand it before you touch the robot

What I learned

FIVE TOPICS

Safety	Working at the cell without putting anyone at risk.
GMAW welding	How wire, gas and an electric arc join the steel.
The WPS	The approved recipe that every weld has to follow.
Defects	Recognizing a bad weld and understanding its cause.
Parameters	The settings that shape the weld, and their limits.

HOW I LEARNED IT

616

pages of training manual and customer standard, read page by page.

Daily

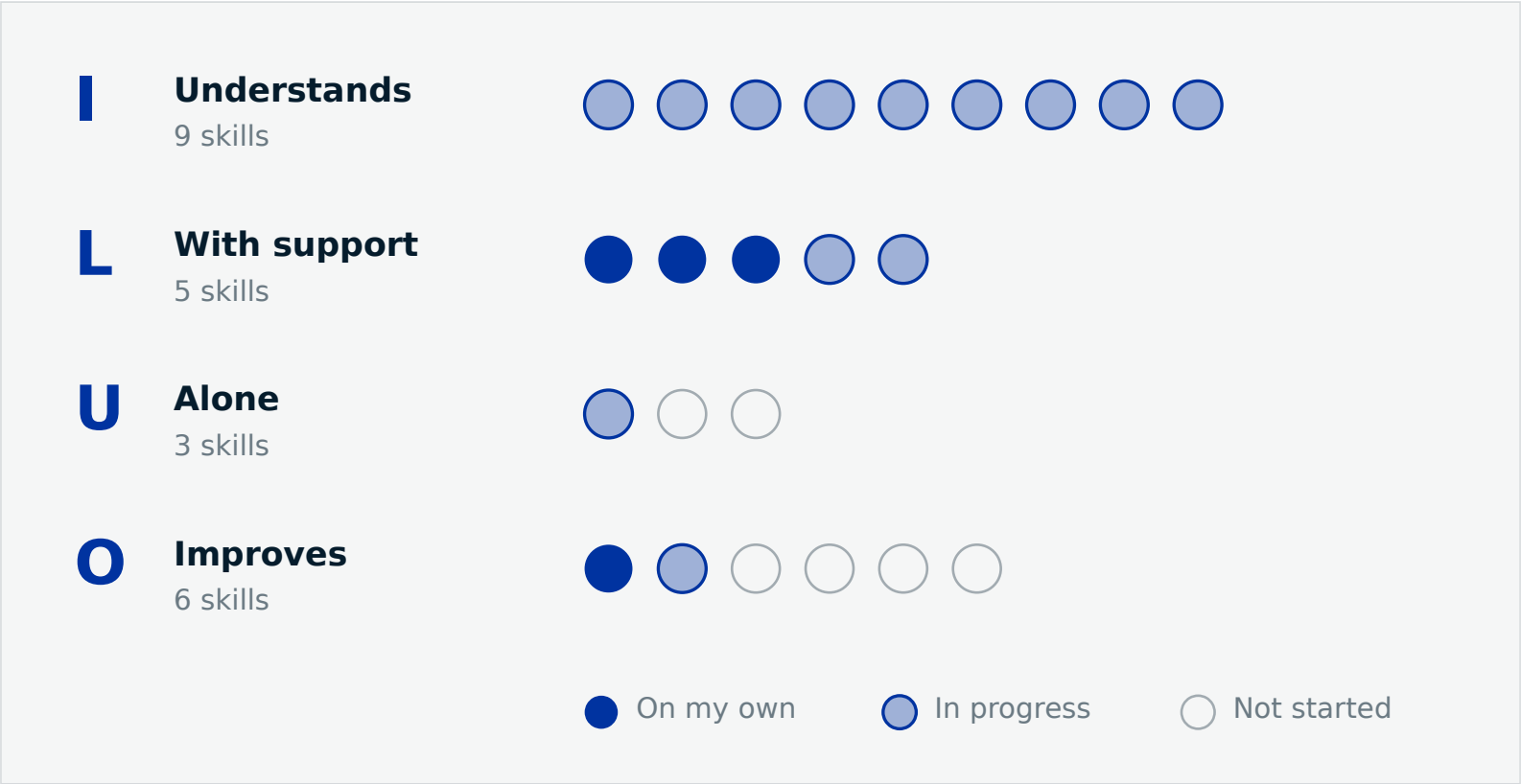
at the cells, attending robot stops with the technicians.

25

questions of a practice exam from previous years, solved.

Four skills on my own, thirteen in progress

Progress on the 23 skills of the program



ON MY OWN TODAY

➤ Changing consumables

➤ Adjusting the key welding settings

➤ Reading weld symbols on drawings

➤ Closing ANDON calls

Level I: exam still pending.

Most of my time goes to industrialization

How my time is split

60%

Industrialization project



40%

Support at the robotic cells: stops, adjustments and ANDON calls.

THE PROJECT

Industrialization of the new Cummins project, together with Engineering.

WHAT INDUSTRIALIZATION MEANS

Getting a new process ready to run on the floor: equipment, programs, trials and the release to production.



Improvement idea: from a manual booth to the 4M cell

Improvement idea and implementation | Done

THE PROBLEM

Long welds, done by hand in a manual booth: 146 s of welding per part, in uncomfortable positions.

THE IDEA

Move the process to 4M, an automatic robotic cell, with the same fixture.

IMPLEMENTED

Running in 4M today. The robot welds, and the team member only loads and unloads.

PHOTO

Before: manual welding booth

PHOTO

After: 4M robotic cell

Same part, 47% less cycle time

Improvement | Results, in seconds per part

CYCLE TIME PER PART

Manual booth



4M robotic cell



■ Load and unload ■ Welding by hand ■ Welding by robot

Same fixture, so loading and unloading take the same time. The gain is all in the welding.

-47%

cycle time, 174 s to 93 s

-56%

welding time, 146 s to 65 s

81 s

saved on every part

What I would change in how setters learn

Final project | Training process update, a proposal

01

Reviewed

616 pages of training manual and customer standard, page by page, against what Level I asks.

02

Found

Gaps: topics the exam asks about that the material barely covers, and an error in one key table.

03

Rebuilt

The schedule, in the plant's own format, so plan and reality can be compared week by week.

PROPOSAL

Use the corrected material and the new schedule with the next group of setters.

Next: from knowing it to doing it alone

What comes next, until January

NEXT LEVELS

L Validate settings and find defects at the cell, with support.
November

U Create and set up robot programs on my own (SKS).
December

O Solve problems, share what I learn, keep improving.
Already started with 4M

WHAT I NEED

- A date for my Level I exam
- Supervised time programming robots in SKS
- Your feedback on the training proposal



Conclusions and next steps

What is in place, and what comes next

IN PLACE

Level I studied in full, and four skills done on my own.

A manual weld moved to the 4M robotic cell: 47% less cycle time, less strain.

A reviewed study path, and a schedule that shows plan versus reality.

NEXT

Pass the Level I exam in October.

Reach Levels L and U before January.

Keep measuring 4M, and find the next weld to automate.

I came to learn robotic welding, and I am already improving it.

The background of the slide is a dark blue image of a tunnel. Multiple bright light trails, resembling headlights or laser beams, streak across the frame from the top left towards the bottom right, creating a sense of motion and depth. The perspective is from the entrance of the tunnel, looking down its length.

TENNECO

Thank you

Diego Adair de León Márquez | Engineering Department